

Smart Renewable Microgrid

A community-scale system that combines renewable generation, battery storage and smart demand management.

7

AFFORDABLE
& CLEAN ENERGY

01 · PROBLEM

655M

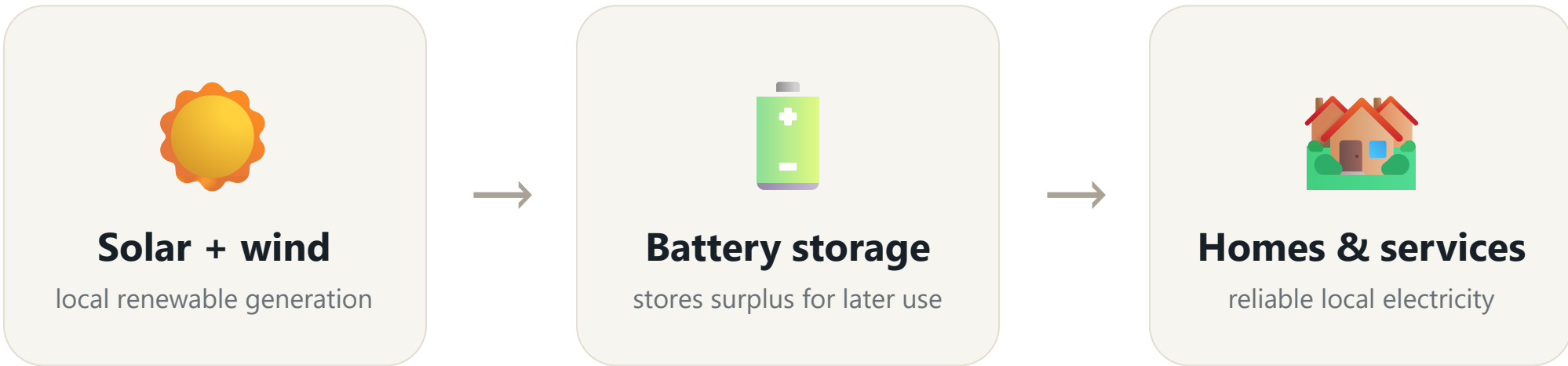
people still lack access to electricity globally

- Access & affordability**
Remote and vulnerable communities remain among the hardest to reach with reliable electricity.
- 2 billion people**
still use polluting fuels and technologies for cooking.

02 · PROPOSED SOLUTION

Local clean power + storage + smart demand

The prototype produces electricity close to users, stores surplus energy and helps households shift flexible consumption away from peak hours.



SMART ENERGY APP

Shows current consumption and suggests cheaper times to use flexible appliances.

78%

Battery level

22:00

Suggested off-peak use

04 · EXPECTED IMPACT

What changes if the prototype is implemented?

- More reliable access**
Local generation and storage can support continuity of supply.
- Lower emissions**
More renewable electricity can reduce fossil-fuel dependence.
- Better affordability**
Efficiency and demand shifting can reduce pressure during peak hours.
- Skills & jobs**
Installation, maintenance, data and grid-management roles.

03 · STAKEHOLDERS

- Government & regulators
- Grid and energy companies
- Investors & technology providers
- Local communities & households

RESEARCH CONNECTION

SDG 7 tracking identifies distributed clean-energy solutions, stronger grids, investment and policy support as important parts of expanding reliable energy access. This prototype combines those ideas in one community-scale system.

KAZAKHSTAN CONTEXT

6.2%

of electricity production came from renewable sources in 2024, excluding large hydropower. This gives the prototype a clear local connection without limiting it to Kazakhstan.